

Data-driven performance management of business units using process mining and DEA: case study of an Iranian chain store

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Abstract

Purpose – The underlying purpose of this paper is to propose a comprehensive framework evaluating the performance of business units of an organization with a process perspective, identifying the most influential performance indicators, enabling managers to make more informed decisions based on data recording every day in their operational information systems.

Design/methodology/approach – For proposing the conceptual framework of performance evaluation a synchronized analysis of selected process' data, obtained from an integrated information system of an Iranian chain store, was performed.

Findings – The superiority of the proposed framework results is demonstrated in comparison to applying the process mining solely; principal component analysis was identified as an efficient link between process mining and data envelopment analysis. Also, based on the final data analytics, the units' throughput times and the variety of brands and suppliers had the most impact on their performances.

Research limitations/implications – The data of abundant business units and performance indicators, which would have allowed adding data prediction and other data analytics techniques for more insight, was not able to be accessed.

Practical implications – Organizations' managers can use the framework to evaluate their business units' current status and then prioritize their resources based on the most influential performance indicators for overall improvement.

Originality/value – The study contributes to the research on performance management and process mining by presenting a comprehensive framework with two levels of data analytics. It stresses discovering what is happening in business units, and how to prioritize their improvement opportunities learning the significant correlations between performance indicators and units' performance.

Keywords Performance management, Process mining, Data analysis, Business process management, Supply chain management

Paper type Research paper

1. Introduction

It is undeniable that managing an organization requires an integrated knowledge extraction out of all of their business units. For example, a distribution part of a supply chain (SC) handles a complicated process (Lambert and Cooper, 2000) whose output directly depends on the performance of various parts; the management level of each SC needs to be able to continuously analyze their units' processes under their control from various aspects (Croxtton *et al.*, 2001); the result of this analysis relies on the amount of their as-is process awareness.

Business process management (BPM) is a management discipline that identifies and manages the business processes of an organization, and the aim of its application is an improvement (van der Aalst *et al.*, 2003; Dumas *et al.*, 2013; White and Cicmil, 2016). Nevertheless, nowadays, paying attention to the data of information systems



reporting a process is much more reliable for discovering what is happening in an organization than BPM data gatherings, such as subjective interviews and sampling. Although BPM techniques are still required for defining the to-be process and facilitating process discovery.

Process mining (PM) is a data-driven approach to reconstruct, analyze and improve business processes using log data recorded by information systems, bridging the gap between model-based process analysis (such as simulation and BPM techniques) and data-centric analysis techniques (such as data mining and machine learning) (van der Aalst, 2016). In general, by using PM, one can discover the processes, recognize bottlenecks of different process paths, check the as-is processes conformance in comparison to to-be ones and provide suggestions for enhancing the existing process (van der Aalst *et al.*, 2011). PM has been successfully applied in different fields, such as healthcare, information technology, manufacturing, education, finance and logistics (dos Santos Garcia *et al.*, 2019). Substantial applications of PM have been executed in hospitals because of the complication of healthcare processes (Badakhshan and Alibabaei, 2020). Moreover, PM's time-efficiency execution has been also proved as a way of analyzing and improving processes (dos Santos Garcia *et al.*, 2019).

The innovation of this paper is primarily proposing a comprehensive performance management framework with accurate, no-limited-detailed process discovery, with two levels of data analysis, which will lead to a better understanding of defects, informed decision-making and an effective overall improvement.

For presenting the defined framework application in business, this study was carried out by case study of Shahrvand Chain Stores Company in Tehran and with the help of the academy team of Celonis, the market-leader provider of PM; its tool has been used for analyzing the process and programming custom dashboards. First, the as-is process of satisfying customer needs in distribution units, using event logs prepared from the raw data of Shahrvand's information systems, is discovered. Then the evaluating phase starts; a dashboard is designed to measure process performance indicators (PIs) so that managers in an SC could have the PI results of business units by importing the data related to their desired time interval. Then, the dashboard results were summarized by principal component analysis (PCA) to provide suitable input for data envelopment analysis (DEA) which is an efficiency measurement technique. After that, the stores were ranked by running one of the DEA models in DEA-solver, and the improvement potentials were identified using data science techniques using Python programming language. By data visualization, a better comparison of units is achievable, and also, it has been presented which PIs were the most influential ones in gaining a high rank among different units. Besides, the comparison of results between using the proposed framework and using only PM is also mentioned.

The remainder of the paper is organized into six sections. The next section will provide a literature overview and related key concepts. In the third section, the defined framework of this study is explained. The fourth section shows how to implement the framework in the case study. In the fifth section, the paper's main results, in the sixth section, the discussion, and finally, in the seventh section, the conclusions of this study and the future study suggestions are presented.

2. Literature review

A business process is a set of activities that generates customer value. The concept of PM was first introduced by Agrawal *et al.* (1998) and Cook and Wolf (1998). Their dedication to workflow management and machine learning led to the development of a new topic called PM. Although PM is a relatively young discipline, there is a growing trend of its application in both scientific and business aspects (Kerremans, 2018; Vom Brocke and Mendling, 2018).

PM starts with an event log which is a record of what exactly happens when a process is performed. PM applies algorithms on the event log data to uncover the actual process execution with full transparency. The event log should contain three main parts: “case ID” is the ID that allows us to trace every single execution of the process; “activity” describes the process steps being performed; “timestamp” indicates when each of the events has taken place. Each unique execution of the process is called a “variant”, and “throughput time” is the average duration between two different steps of the processes, or even the first time and the last time one single step occurs.

PM can be divided into three main parts: process discovery, conformance checking and process enhancement (van der Aalst, 2011). Process discovery allows process models to be extracted from importing event logs. Conformance checking examines the deviations from comparing a process model with an imported event diagram to determine whether the event and model diagrams fit together (Rozinat and van der Aalst, 2008). The enhancement also allows the existing process model to be improved (van der Aalst *et al.*, 2011). Many research areas have benefited PM in these three main parts (Dakic *et al.*, 2018), as well as the SC research field (Jokonowo *et al.*, 2018), but there is a lack of studies using this technique to evaluate, compare and enhance the performance of various business units.

This section includes articles published in journals and books that have applied PM to performance measurements. Park *et al.* used PM as an auxiliary tool of performance measurement methodology for the first time (Park *et al.*, 2014). They compared and ranked 19 block manufacturing processes in a Korean shipbuilding company by using DEA. They used automated PM results only to measure five basic PIs, including three time indicators, the number of process tasks and the number of raw materials. Later that year, Leyer and Hüttel introduced PM as a performance measurement tool in a book chapter, which describes the process of evaluating process performance as a part of the BPM cycle by introducing the PM and DEA, along with a descriptive application example (Leyer and Hüttel, 2018). In 2016, van den Ingh in his master's thesis at the University of Eindhoven measured the performance of different paths of a purchase-to-pay process in an organization. He developed the Devil's quadrangle framework using outputs of one of the PM tools.

Subsequently, some studies in the education field used PM to evaluate students' performance. First, Umer *et al.* benefited PM to improve the student learning experience in massive online open courses (Umer *et al.*, 2017), and then Baykasoğlu *et al.* to evaluate students' performance in their university computer exams (Baykasoğlu *et al.*, 2018). It has also been used in the e-learning self-evaluation of students (Cerezo *et al.*, 2020). Arias *et al.* have stated that there is a growing interest in human-resource allocation within BPM and PM (Arias *et al.*, 2018), but only one article in this field has used PM from a performance perspective. Sophia and Sarno integrated the Factor Evaluation System (FES) methodology and PM to analyze the performance of different jobs (Sophia and Sarno, 2018). Besides, the transportation industry has recently benefited PM for customer experience and its business performance optimization (Rowlson, 2020).

Finally, the healthcare field has utilized PM seeking better performance. Rojas *et al.* analyzed the performance of emergency room episodes to help decision-makers improve the quality of healthcare services (Rojas *et al.*, 2019). Also, Battineni *et al.*, by using a PM case study and extracting data from a hospital information system, evaluated the performance of the system with time indicators available in any PM tool (Battineni *et al.*, 2019). Recently, some articles analyzed the performance of emergency rooms' processes by PM (Antunes *et al.*, 2019; Cho *et al.*, 2020), and also, it has been shown how insightful PM can be in health and safety performance (Pika *et al.*, 2021).

As mentioned so far, only a few studies have used PM to evaluate performance. Their study areas include manufacturing, education, human-resource, transportation and healthcare, while they evaluate different processes of one business unit, different process

paths, students, jobs, emergency rooms and systems. Thus, PM has been widely applied in a single system or organization (Thiede *et al.*, 2018). According to the above discussion, many areas have not benefited from PM so far, and in most studies, only a few fundamental indicators have been considered, and performance is generally outlined; although PM can mine the smallest detail in any process and provide accurate information, the exact comparison of different units cannot be clarified considering the limited ability of a human facing many aspects of performance.

A framework to overcome these shortcomings and fulfill these gaps is presented in this study, which shows how useful PM can be for comparing the performance of a high number of various business units, and in particular, for managing distribution units of an SC. Defining appropriate performance criteria (by the usage of various systems or an integrated system containing all units' data) can provide precise control for a management level. Also, the utility of PM techniques for calculating these criteria is a suitable way to monitor all performance changes in the shortest possible time. The DEA-Solver has also been used as an auxiliary tool to provide accurate ratings of these units, providing a functional integration with PM. It is self-evident that based on the definition of SC Management (SCM), this research is not only a subset of BPM and PM but also is a subset of SCM since it offers a new, agile way of managing different business units of an SC.

3. Integrated conceptual framework of PM and DEA

The performance management framework proposed in this research is data-driven with the process perspective of various business units of any organization. The basic framework is summarized in Figure 1.

To evaluate the performance of each unit, achieving a precise cognition of its as-is process must be taken as the first and most necessary phase; In other words, evaluation is not rational without enough recognition. For this purpose, the initial steps are defined to understand the

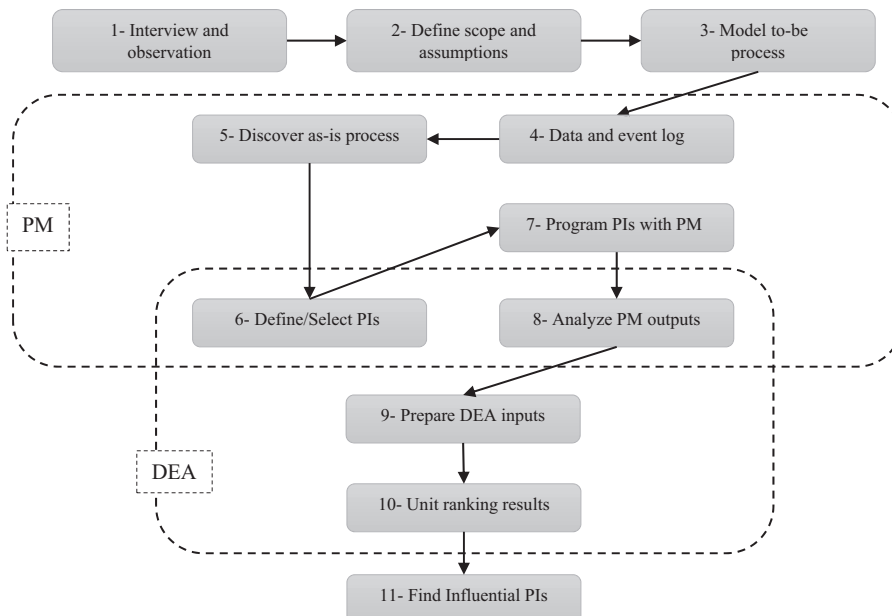


Figure 1.
The proposed
framework of
performance
evaluation

evaluated environment. As the first step, observing the activities of the selected process in the environment and interviewing related managers or staff provides the necessary information about the process. Then in the second step, the scope of the organization for evaluation, process activities, the target timeframe and the assumptions should be defined precisely.

As the third step, from the knowledge gained so far, the to-be process can be modeled by using one of the business process modeling and notation (BPMN) tools, in which activities are shown in rectangles, connectors in arrows and multi-output states are represented in rhombuses (Zur Muehlen and Recker, 2013). This process model can help IT experts to understand process boundaries and the activities required to export raw data from information systems.

As the fourth step, using the designed BPM model, the raw data related to the time log and any other related data of the process under study can be extracted. Since the input data of all PM tools called event log must contain a case ID (to track process paths) and timestamp (to detect activity sequences) of activities (van der Aalst *et al.*, 2004), the raw data received should be preprocessed. More analysis could be obtained by PM if additional features of each event, as the number of goods or costs, were accessible.

In the fifth step, the process is discovered by PM. Discovery can be done separately for each unit or aggregated for the whole organization. Defining the performance indicators (PIs) in the sixth step, first, a proper understanding of the process under study and utilizing several articles in the area of PIs of an organization, especially an SC as the case study is required (Chae, 2009). The final PIs are selected after checking the following conditions:

- (1) Survey of process specialists
- (2) Computable by the available data
- (3) Programmable in the selected PM tool

One can consider an unlimited number of PIs as long as they adhere to the above conditions; In our case study, while the selected PM tool is Celonis uses Process Query Language (PQL), which is “some how” an addition of Structured Query Language (SQL), any concept in one’s mind that can be formulated to a PI, can be programmed in it, so the third condition is mostly not a constraint in this case. Therefore, one of the advantages of using PM techniques for performance management is the feasibility of benefiting unlimited PIs using real-time data in digitalized processes.

It should be noted that the two main differences between Celonis PQL and SQL are PQL does not support all operators that are available in SQL, but, in contrast to SQL, PQL is domain-specific and offers a wide range of PM operators not available in SQL. Consequently, process-based PIs, the most important of which are throughput times, can be evaluated using PQL.

As the seventh step of the framework, the PM technique will be used for measuring the performance of the units. Starting with the eighth step, the analysis phase begins. Practically, efficiency is the ratio of outputs to inputs. A developed model for efficiency measurement is called DEA, which is used as a non-parametric method to calculate the performance of the decision-making units (DMUs) in this paper. Charles, Cooper and Rhodes developed Farrell’s original approach of DEA (Farrell, 1957) and presented a model capable of measuring the efficiency of DMU by simultaneously considering multiple inputs and outputs (Charnes *et al.*, 1994). Today, many theoretical and practical developments have occurred in DEA models, and all the well-known basic models are available to use in software named DEA-Solver (Cooper *et al.*, 2006).

The results of new dashboards developed in PM software are obtained at the eighth step and designated as DEA-Solver tool inputs, but before that, in step nine, the results of

dashboards need to be summarized to a fewer number by PCA and be normalized (Ueda and Hoshiai, 1997). As the tenth step, the units' relative efficiencies and a ranking of units are obtained from DEA-Solver. Finally, as the last step, data analysis between PI values of every unit and the relative efficiencies needs to be performed; this way, by visualizing the Spearman's correlation (statistical dependence between the rankings of two variables), the unknown relationships between the influential PIs and the efficiency of the units will be revealed. Consequently, the management level of an organization will realize their most effective strengths and weaknesses, in order to improve the performance of the units accurately, based on real data, with high transparency, and in the shortest possible time.

The proposed framework allows the management level to update the results in the short term with real-time data, observe the changes quickly across the organization and make any improvement decisions accordingly.

4. Application of the proposed framework on a case study

4.1 Case study introduction (by interview and observation)

Shahrvand Chain Stores Company was established in 1994, and by now, it owns 44 branches in Iran. These stores include five product groups: grocery, protein, clothing, hygiene and cosmetics and home appliances. A study on raw data received from the IT department of Shahrvand is a 3-month report of 2019 from hygiene group turnover at 21 selected branches in Tehran.

There are two levels of data analysis in the framework, which are presented in Figure 2. First, the raw data stored in the organization's systems is preprocessed for preparing the event log (PM input), whereby both data visualizations of the process map and variants, and the PIs outputs have resulted from PM techniques such as PQL. Then, the PIs outputs are preprocessed, reduced in dimension, visualized and utilized for measuring efficiencies and ranking the units by DEA. As the second main level, a correlation test has been applied on the data frame containing both the PI values and efficiency scores of all the business units to find the most influential ones. This way the decision-makers can manage the enhancement actions or projects of weak units.

4.2 The scope and assumptions of the process

In the beginning, during a meeting with Shahrvand's managers, general information was gathered, such as the approach of order, the steps of delivery from suppliers, warehousing and selling goods. The scope of the SC, process and activities studied are shown in Figure 3.

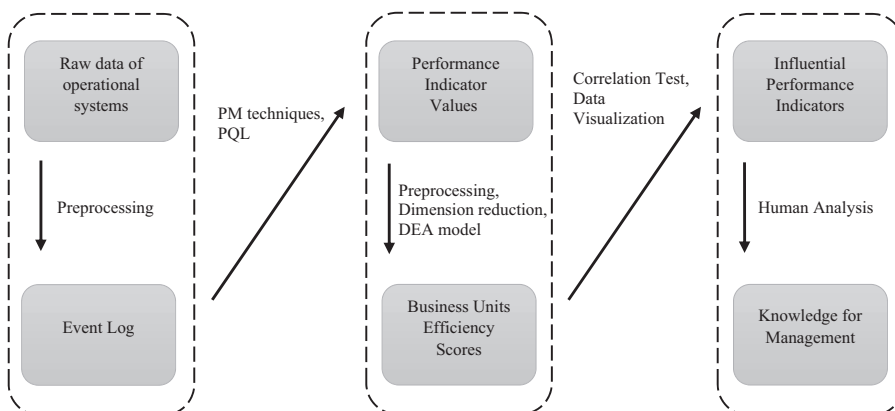


Figure 2.
The levels of data analysis

The assumptions of this modeling include the following considerations that are required in both BPMN and PM results:

- (1) Each order = each process case: In this research, every time order of batches is placed, a process case has started, and in the same way, every order ID is considered as a case ID. For clarification, because of the system's limitation, each process case contains each sales batch per day, traced backward in each order.
- (2) The order-to-sales process involves numerous activities or sub-processes: Several other steps are within the ones mentioned in [Figure 2](#), which have been overlooked due to the absence of time logs in the existing information system. Some of the steps are better to be called sub-processes, but they are referred to as activities due to the research limitation.
- (3) Return from customer (C) = Return to sales level. Although it is more reasonable to examine the product returned and if there is any defect or problem with it to be returned to the supplier, because of the inaccessible details of the returned goods, it has been assumed that the product is returned to the shelf, or the process has ended.
- (4) Returning to supplier (S) reasons: There is no distinction of referral types in the system, for instance, skipping the expiration date, damaging or not being sold.
- (5) First entry to shelf, first sale or first-in, first-out (FIFO): Due to the invisibility of the goods after entering the shelves, the study assumes that the goods are sold respectively.

4.3 Process modeling by BPMN

The following model was obtained by interviewing the planning manager, IT manager, ordering expert and also observing the order-to-sales process at one of the stores. By using Visual-Paradigm modeling software, the process diagram is presented in [Figure 4](#).

4.4 Extracting raw data and preparing event log

Preparing an event log from the raw data provided by Shahrvand's integrated information system challenged the research process because it is designed to show the activities taken place per day, but what is needed is the characteristics of every activity, including its date of occurrence. Some rows of the Excel file reported from this system are given as an example in [Table 1](#).

A sample event log, extracted from the raw data of winter 2019, from Shahrvand's information system, is shown in [Table 2](#).

4.5 Process discovery

The PM tool used in this paper is called Celonis. This research is performed at the analyst level users of Celonis and its valuable data visualizations is presented in this section.

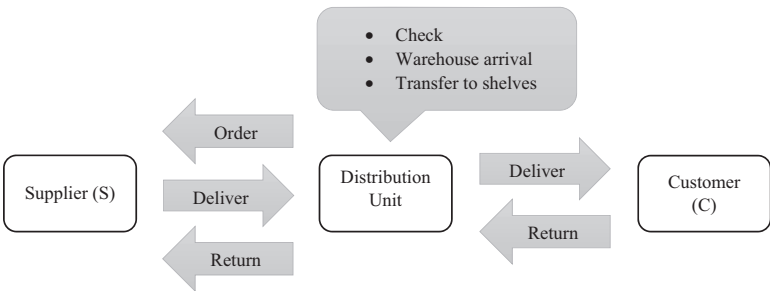


Figure 3.
The scope of supply
chain studied

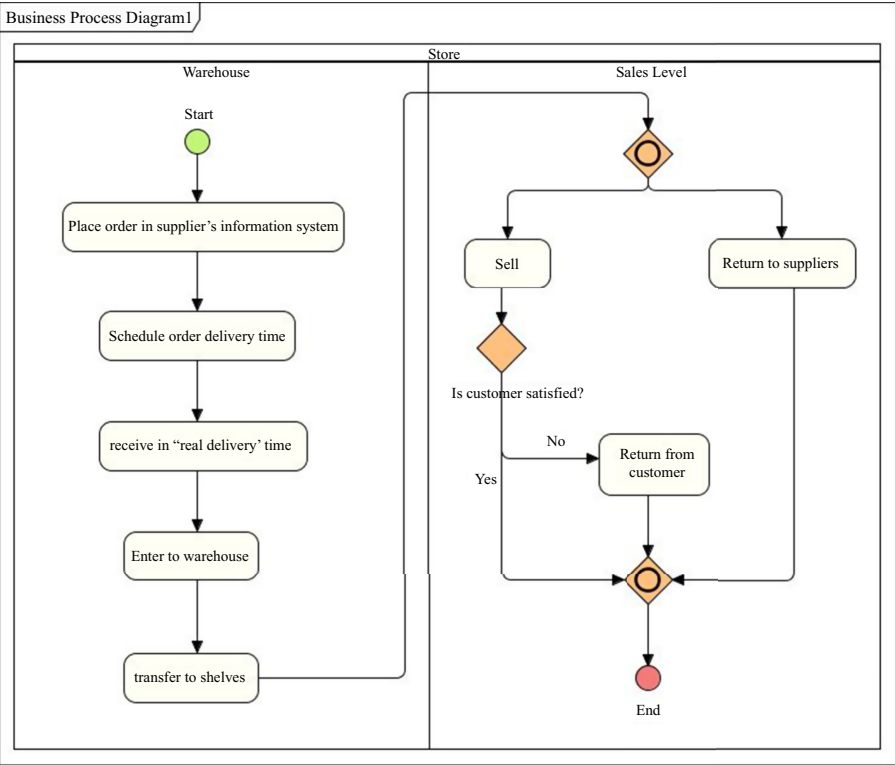


Figure 4.
Business process
diagram

Date	Store	Barcode	Supplier	Goods description	Ordering ID	Schedule delivery	#orders	Delivery ID	#delivery	Transfer ID	#transfers	Return to S ID	# return to S	#sales	#return from C
2018 -23 -12	#3	763002	Gum	A 75ml	9728768	2018 -30 -12	72								
2018 -30 -12	#3	763002	Gum	A75ml				982876	72	982876	72	456588	12	2	0

Table 1.
Sample of data
extracted from
Shahrvand
information system

4.5.1 *Process map.* Figures 5 and 6 are the diagrams of the selected process with maximum detail in all units. The significant difference of the “as-is” process map based on actual data with the “to-be” BPMN model presented in Section 4.3 is undeniable. The numbers in Figure 4 indicate the number of cases involved in each activity (node) or connection (arc), and the numbers in Figure 5 indicate the duration spent on them. As can be seen, the process begins with 183 times batch order.

According to Figure 5, the process includes a sample of 183 ordering cases. Once the order has been recorded, the scheduling time in the supplier system is determined. In 177 cases, after the scheduled delivery date, the goods are received, but in the remaining six cases, the supplier delivers the goods earlier than the scheduled delivery time. In 55 cases, the goods are not received from the supplier, and in 128 cases, the goods are received, checked, imported into the warehouse and finally transferred to the shelves for sale. In this step, the goods are either sold or returned to their suppliers. After selling goods, there were four situations. In four cases, some other goods in the same batch were returned to their suppliers. In six cases, the supplier had sent an incomplete amount of order, so the process goes to a “no delivery” state eventually. In the third situation, there was a “return from customers” in 14 cases. Finally, in remained cases, the process had ended after the customer has bought the product. The color shades presented in this map demonstrate the frequency of cases passing through each activity or connection; lighter colors mean less and darker ones mean more passing.

Figure 6 can be analyzed accordingly; the numbers on the connections show the average duration of cases spending in each one. They can be altered to median duration, in hours and minutes. Again, the lighter colors show less time being spent on a connection, and darker colors might show bottlenecks of the process.

4.5.2 *Process variants.* After importing the event log prepared in Celonis and using the software’s discovery algorithm, the process map is designed with the selected detail mode in the most understandable way possible. In general, the order-to-sales process at 21 branches of Shahrvand store during winter 2019 comprises 16 unique different variants. Figure 7 illustrates the outputs of the five most coverage paths of the process.

The first variant shows the happy path of the process, with 52% of total coverage in 96 cases. The second variant, with 30% coverage, shows the supplier has missed delivery in 55 cases, after ordering and scheduling the delivery time. The third variant, with 4% coverage, shows customers had returned some goods. The fourth variant, with 3% coverage, indicates the processes in which the real delivery was sooner than the scheduled delivery. The last path presented, with 2% coverage, shows the incomplete deliveries from suppliers in four cases.

4.6 *Define/select PIs*

With the explanations in mind from section 3, process PIs were defined. The undesirable PIs, which their minimization is beneficial, are inputs; the desirable PIs, which the organization will seek to their maximization, are assigned to outputs of the process. For more clarification, these PIs are defined based on their direct or reverse impact on what makes value for the business units of our case study; these can discover four basic categories of improvement opportunities in our case: process steps (desirable/undesirable activities, such as cases with/without a specific activity), sequence (the order of activities, such as happy path coverage),

Table 2.
Sample of event log

Activity	Timestamp	Case-ID	#Goods	Brand	Store
Order	2018-23-12	982876	72	A	#3
Schedule delivery	2018-23-12	982876	72	A	#3
Real delivery	2018-30-12	982876	72	A	#3

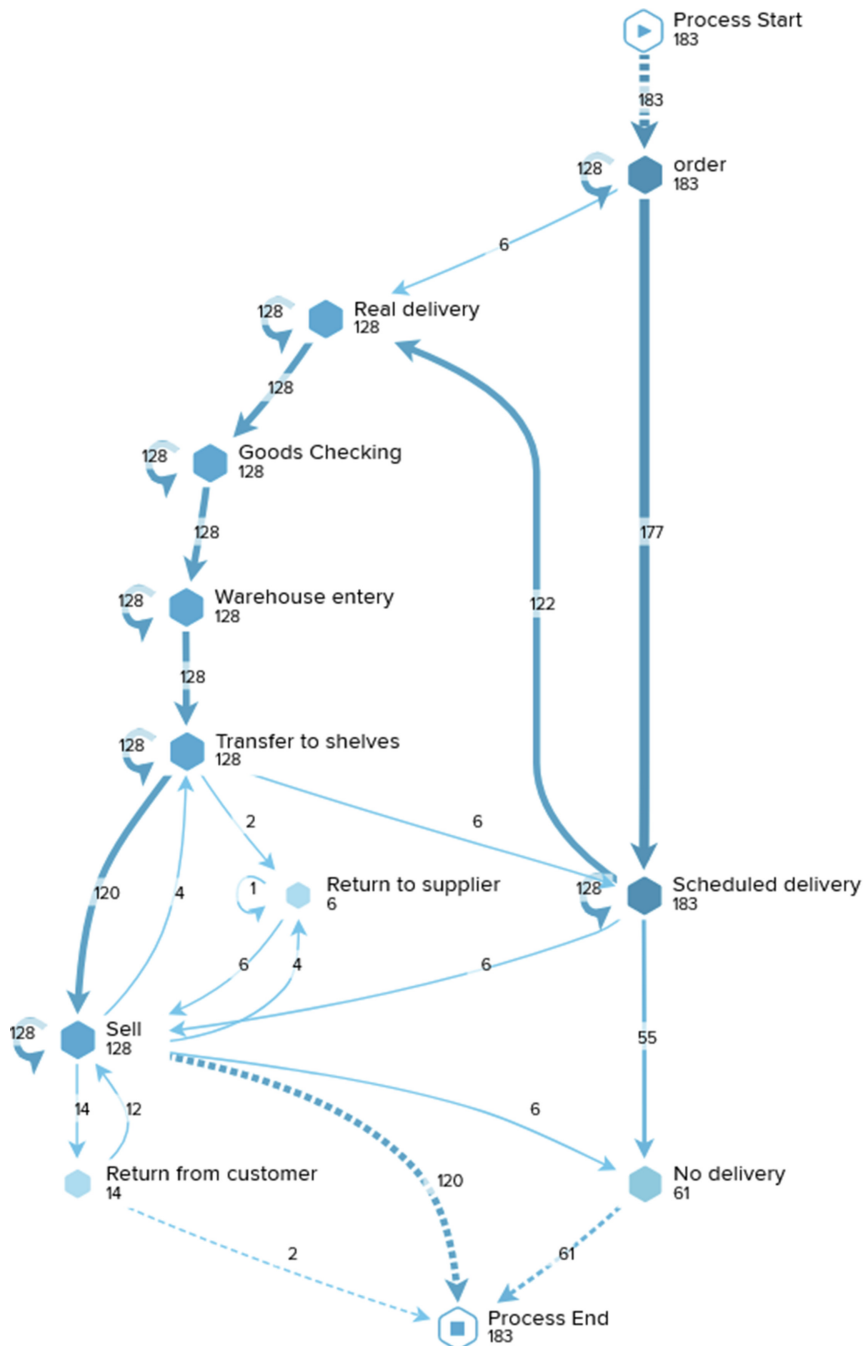


Figure 5.
Process map 1 (case
frequency)

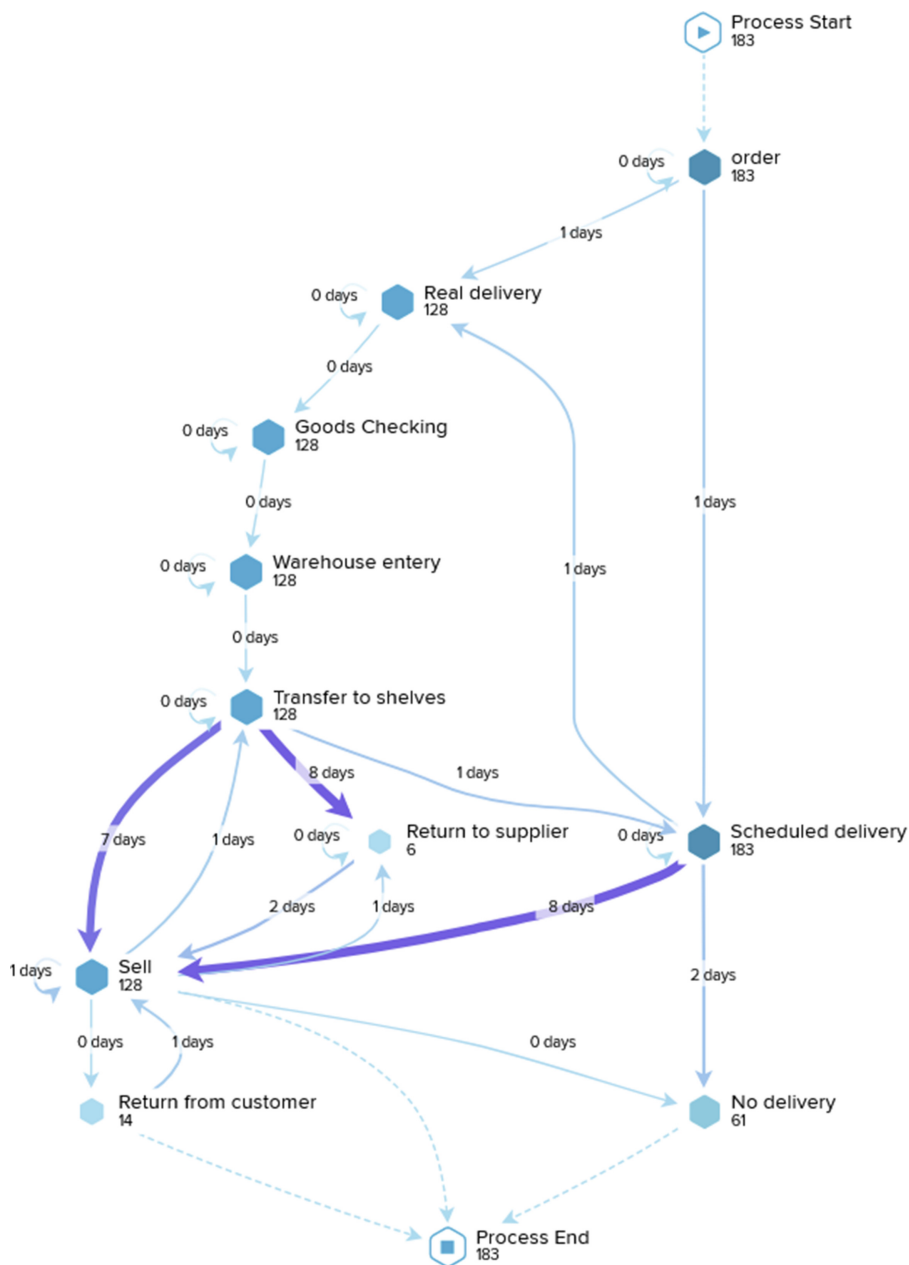


Figure 6.
Process map 2 (average
duration)

time (average duration between different events) and attributes of each case (such as sales amount, return amount, brand variety, etc.). PIs and their mean, minimum and maximum amount for the 183 cases are presented in [Tables 3 and 4](#), respectively.

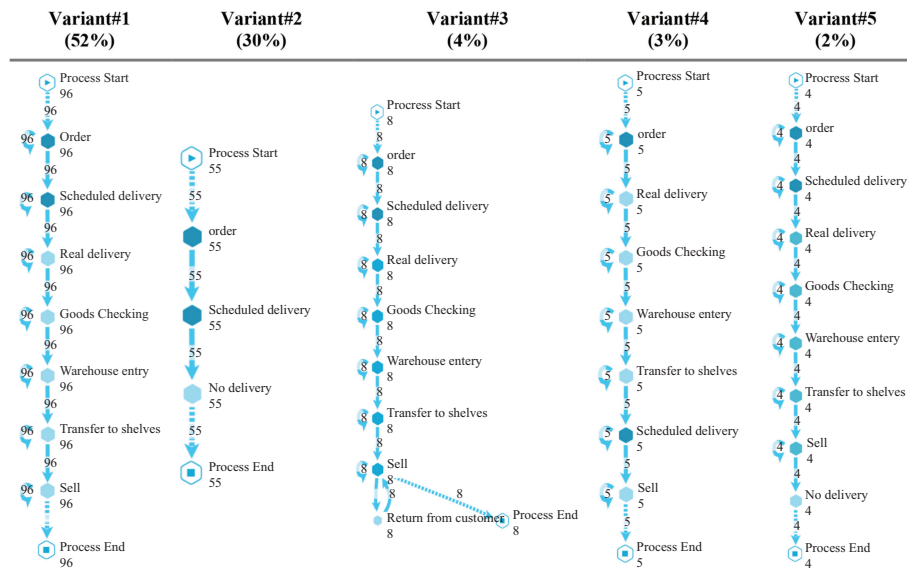


Figure 7.
Process variants

#	Input criteria	Input PIs	Mean	Min	Max
1	Minimum process throughput times (days)	1. Average scheduled delivery duration 2. Lead-time (order-to-delivery) 3. Average supplier delay 4. Average total process duration 5. Average total duration without waiting time 6. Maximum chain-cycle time 7. Average order-to-sales duration 8. Average first-to-last sales duration 9. Average delivery-to-return(S) duration 10. Average delivery-to-sales(first) duration	0.38 1.19 0.38 42.95 41.76 58.81 43.33 36.19 7.71 5.14	0.00 0.00 0.00 17.00 17.00 20.00 17.00 8.00 0.00 1.00	2.00 4.00 3.00 76.00 76.00 88.00 78.00 75.00 68.00 14.00
2	Minimum order and inventory risks	11. % ordering-rework 12. % incomplete-delivery 13. Average ordering times to different suppliers 14. % no-sales-days	0.30 0.03 1.14 0.32	0.00 0.00 1.00 0.06	0.71 0.33 2.00 0.84
3	Minimum return to suppliers	15. % Cases with return(S)	0.04	0.00	0.34
4	Minimum customer dissatisfaction	16. % Cases with customer return	0.13	0.00	0.53

Table 3.
Input performance
criteria and PIs

4.7 Program PIs in PM tool

All PIs defined should be coded in the new dashboards of Celonis. The programming language for creating new dashboards in Celonis, PQL, is used for the performance evaluation phase; this allows achieving the result of defined PIs in a short period by importing any new event log of any new relevant process.

Table 4.
Output performance
criteria and PIs

#	Output criteria	Output PIs	Mean	Min	Max
1	Maximum order and inventory ability	1. Average order volume	217.85	63.00	467.00
		2. Overall received volume	1217.04	383.00	3374.00
		3. Average received volume per-supplier	246.04	61.00	626.00
2	Maximum sales volume	4. Overall sales volume	1211.28	383.00	3369.00
		5. Average sales volume per day	25.61	4.00	89.00
		6. Average sales volume per-supplier	245.09	61.00	626.00
3	Improved supplier relationships	7. Maximum purchase share among suppliers	42.28	21.00	72.00
		8. % happy path coverage	0.54	0.33	0.80
4	Maximum happy path adherence	9. % brand-variety	0.75	0.54	1.00
		10. % supplier-variety	0.92	0.69	1.00

4.8 PIs results

After obtaining the outputs of dashboards designed in Celonis and separating the input and output PIs, the results are presented in the following plots. It should be mentioned that for better understanding of the visualized PIs results and easier units' comparing, they are scaled with "minmax_scale" Python approach.

4.8.1 *Input PIs.* The results of input PIs, gained from new dashboards programmed in Celonis, are presented in this section.

It can be seen from Figure 8 that units 3, 4, 5, 18 and 20 have the shortest durations between different activities, so in other words, these are the fastest units.

The results of the order and inventory criteria in Figure 9 indicate that the performance of units slightly differs from different PI perspectives, so the comparison of this criterion is not easy to be analyzed visually.

It can be deduced from Figure 10 that stores 5 to 11, 13 and 15 to 21 were not involved with returning goods to any supplier, and stores 3, 10, 12 to 15, 17, 18 and 20 did not have any customer return.

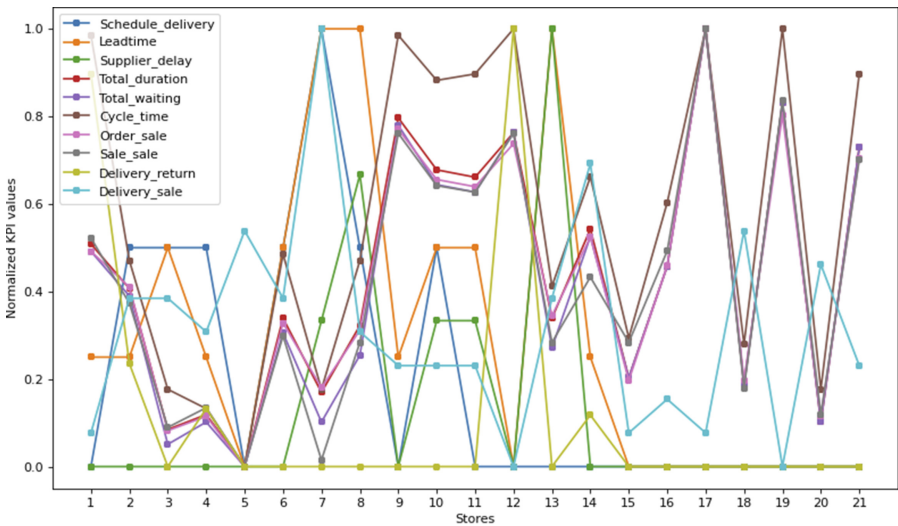


Figure 8.
Throughput times

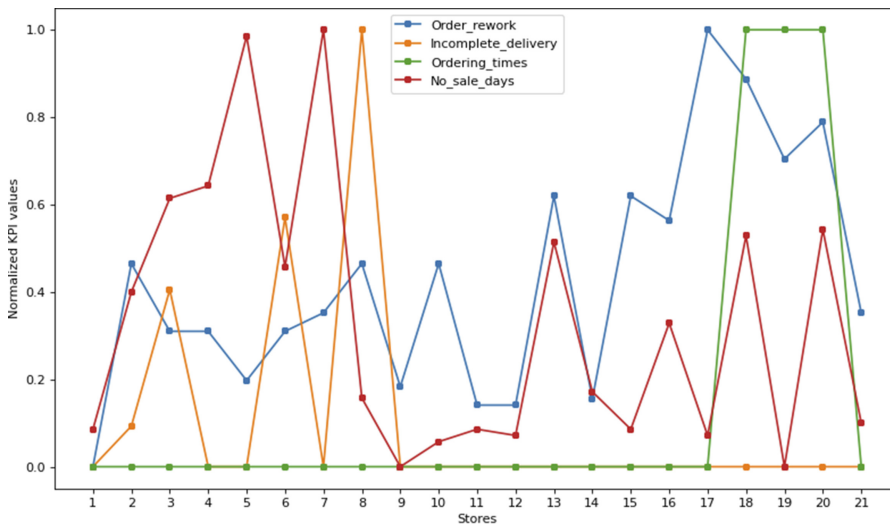


Figure 9.
Order and
inventory risks

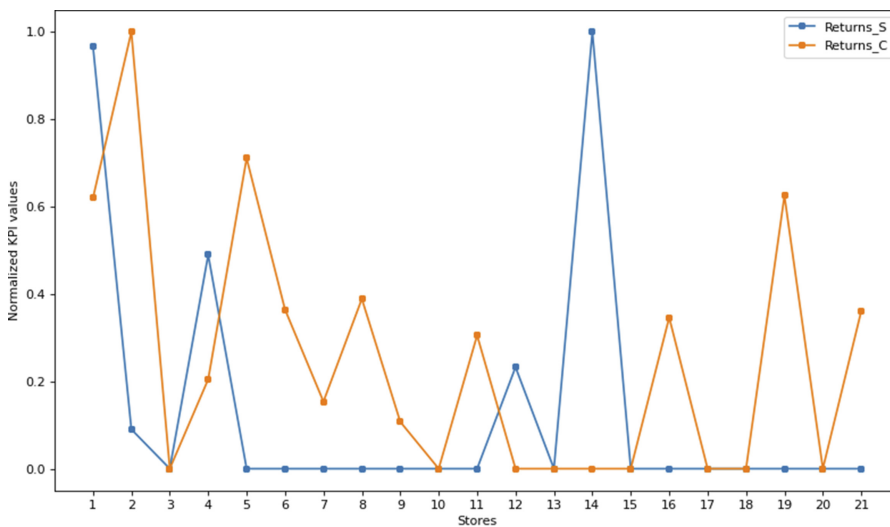


Figure 10.
Goods returning

4.8.2 Output PIs. The results of output PIs, gained from new dashboards programmed in Celonis, are presented in this section. In the diagrams presented, in contrast to input PIs, the optimal level of output PIs is their maximum.

It can be inferred from [Figure 11](#) that in terms of ordering and inventory abilities, stores 1, 3 and 4 performed significantly better than the others.

As can be seen in [Figure 12](#), as in the previous diagram, again stores numbers 1, 3 and 4 performed better in terms of sales.

In [Figure 13](#), the results of the PIs do not provide a specific comparison result. Stores numbers 1, 9 and 11 have followed the happy path of the process. In terms of the two last PIs

Figure 11.
Order and inventory
ability

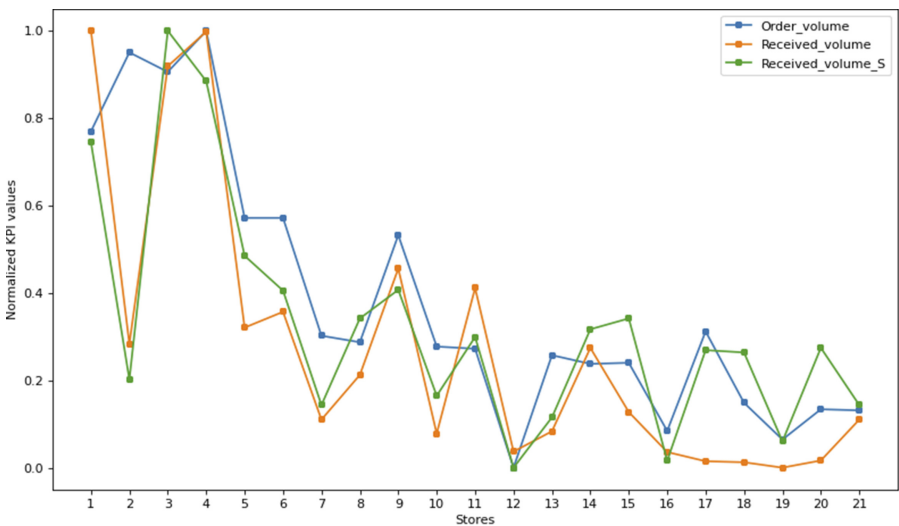
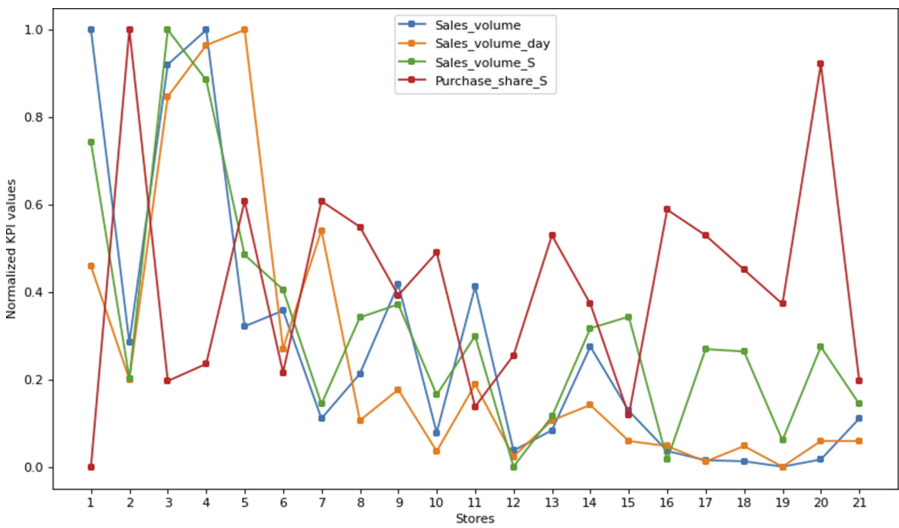


Figure 12.
Sales



(brand and supplier variety), stores 13, 14 and 15 had the maximum brand product and supplier flexibility.

4.9 DEA inputs

The programmed dashboards in Celonis, which results were represented in the previous section, are 26, but according to the proposed DEA-Solver rules, to achieve better ranking results, the number of DMUs should be at least twice the number of input PIs in output PIs. One of the most effective ways to reduce the number of PIs for comparison between multiple DMUs

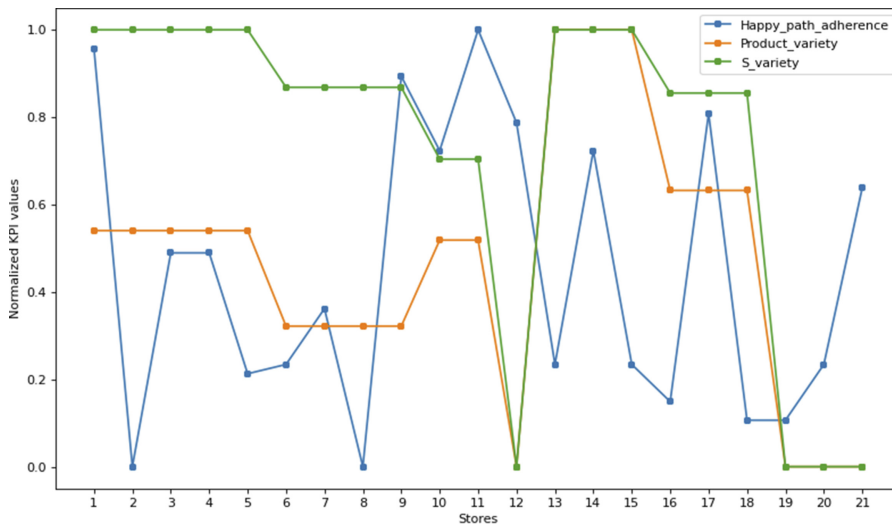


Figure 13.
Happy-path adherence and flexibility

is Principal Component Analysis (PCA). The 26 PIs represented by X_1 to X_{26} , are reduced to a few principal components (Y), which are linear combinations of X_s . The results of the input PIs for finding Principal components using SPSS software are presented in Table 5. It should be noted that the time indicator corresponding to the average time between real delivery and return to the supplier (X_9) was omitted due to the lack of returned goods in many stores.

The number of Principal Components is acceptable when cumulatively over 70% is covered; this article considers the number with at least 75% of the total coverage. For this reason, the first four principal components will be considered among the input PIs, accounting for 80.4% of the total input PIs. Their final results for each store are presented in Appendix 1.

The results of the Principal Components analysis of output PIs were likewise presented using SPSS software in Table 6:

As can be seen, only the first two linear combinations of all the output PIs can account for 75.3% of them. The results of the two Principal Components outputs for each store are presented in Appendix A.

Component	Total	Initial eigenvalues		Extraction sums of squared loadings		
		% Of variance	Cumulative %	Total	% Of variance	Cumulative %
1	6.587	43.910	43.910	6.587	43.910	43.910
2	2.456	16.375	60.286	2.456	16.375	60.286
3	1.871	12.472	72.758	1.871	12.472	72.758
4	1.154	7.693	80.451	1.154	7.693	80.451

Table 5.
Total variance explained (input PIs)

Component	Total	Initial eigenvalues		Extraction sums of squared loadings		
		% Of variance	Cumulative %	Total	% Of variance	Cumulative %
1	5.786	57.857	57.857	5.786	57.857	57.857
2	1.750	17.498	75.354	1.750	17.498	75.354

Table 6.
Total variance explained (output PIs)

Up to this point, with only four linear combinations of input PIs and two linear combinations of output PIs, more than 75% of all 25 is monitored. After normalizing the results of these six principal components, the store performance evaluation process moves on to the next step, ranking by DEA-Solver.

4.10 Units ranking by DEA-Solver

After preparing the final results of the six principal components of the comparison, the values are imported into DEA-Solver ([Appendix 2](#)). Then the “Constant input-oriented Super efficiency Slack-based-measurement (SBM)” model was applied to the data. That is to say, with this choice, in an evaluation based on slacks (SBM concept) and without performance limitation to one (super efficiency), it was first attempted to reduce input sources (input-based concept) as much as possible, and then “exactly to the same rate” (model constancy concept) outputs increase as the second goal (orientation concept). The results of the relative performance and ranking of the DMUs are presented in [Table 7](#).

As can be seen, the distribution units ranked 1 to 21 respectively by their relative efficiency scores. This ranking was based on 25 process-oriented PIs, which was not possible by the utilization of PM alone. Of course, it was possible to compare them with lower precision based on the visualized PI results ([Section 4-8](#)) by Celonis, but with the increasing number of PIs, additional methods such as DEA are needed. Performance ranking of Shahrvand store had not been executed before this study, but according to what managers had in mind, stores 4 and 5 were considered as strong ones, and numbers 11 and 12 were poorly performing ones, which comply with the framework’s results.

4.11 Influential PIs for effective enhancement

In this part, the significant relationships between some PIs and the relative efficiencies of the DMUs are revealed. As a reliable approach, a heat-map of Spearman’s correlation

Rank	DMU	Relative efficiency score
1	18	2.009
2	5	1.782
3	4	1.353
4	13	1.173
5	3	1.079
6	14	1.018
7	20	1
7	7	1
7	1	1
10	15	0.704
11	2	0.690
12	16	0.631
13	17	0.425
14	8	0.369
15	6	0.365
16	9	0.354
17	10	0.338
18	11	0.249
19	19	0.178
20	21	0.073
21	12	0.014

Table 7.
DMUs ranking

between each of the variables is presented in [Figure 14](#). As can be seen, the variables in the data set imported for Python analysis are all of the PIs and the last one is named “Target”, which is the relative efficiency score of the DMUs in our case. The correlations are shown by colors: the lightest and the darkest ones mean +1 and -1, which demonstrate a strong relationship, and the in-between colors mean weaker relationship and even no relationship (when the correlation equals 0) between each pair of variables.

As an assistant to the above analysis, the visualizations of each PI (horizontal axis) based on relative efficiency scores of the DMUs (vertical axis) are presented in [Figure 15](#); so there should be 25 scatter plots for each PI values, and by analyzing them, one can understand whether there is a meaningful relationship between them or not.

Based on the heat-map and the scatter plots presented, it can be concluded that the most correlation between input PIs (undesirable) and efficiency (Target) are: “average total process duration” (-0.76), “average total duration without waiting” (-0.77), “maximum chain cycle time” (-0.77), “average order to sales duration” (-0.75) and “average first-to-last sales duration” (-0.77); the value of these correlations is negative as expected, which means as the PI values increases, the efficiency lowers.

Moreover, a relationship between the efficiency and two input PIs (“average delivery to sales duration” and “%no-sales day”) is realized; however, based on the respective scatter plots, outlier data have influenced this part of the result and is not reliable.

The most significant correlations between the output PIs (desirable) and efficiency are: “average sales per day” (0.51), “brand variety” (0.63) and “supplier variety” (0.68). These positive values indicate a direct relationship between the PIs and the efficiency. The exact value of all of the correlations can be found in [Appendix 3](#).

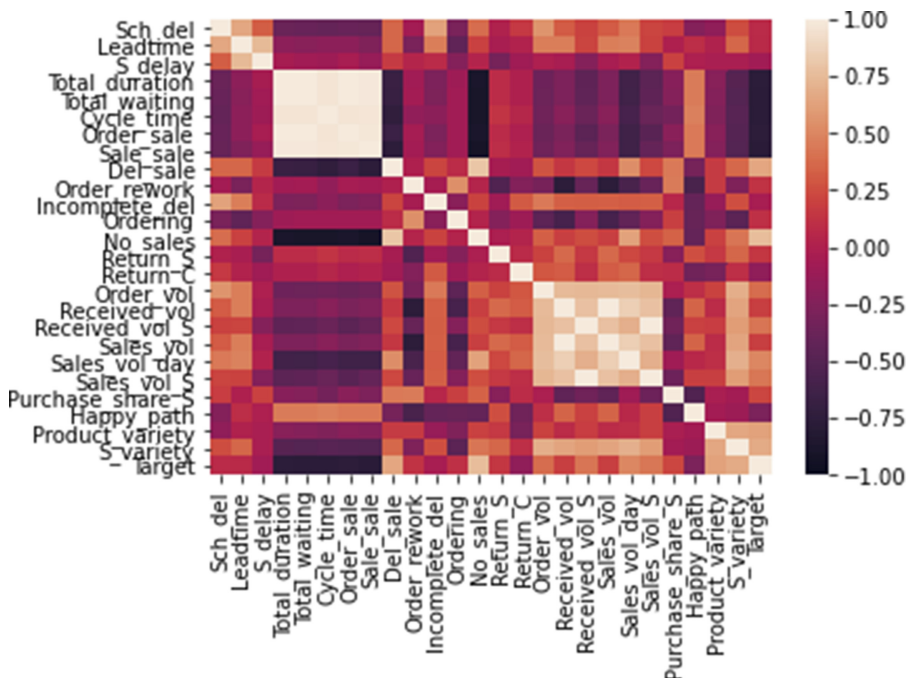


Figure 14.
Heat-map of Spearman
correlation between all
variables

Therefore, this observation is conducive to a significant improvement suggestion for all of the distribution units: their major concentration on shortening their throughput times and growing the rate of their brand variety, supplier variety and after that sales amount.

5. Results

This paper demonstrated how useful a precise cognition was in performance evaluation, and PM is able to penetrate any desirable aspect of performance and plays a primary role

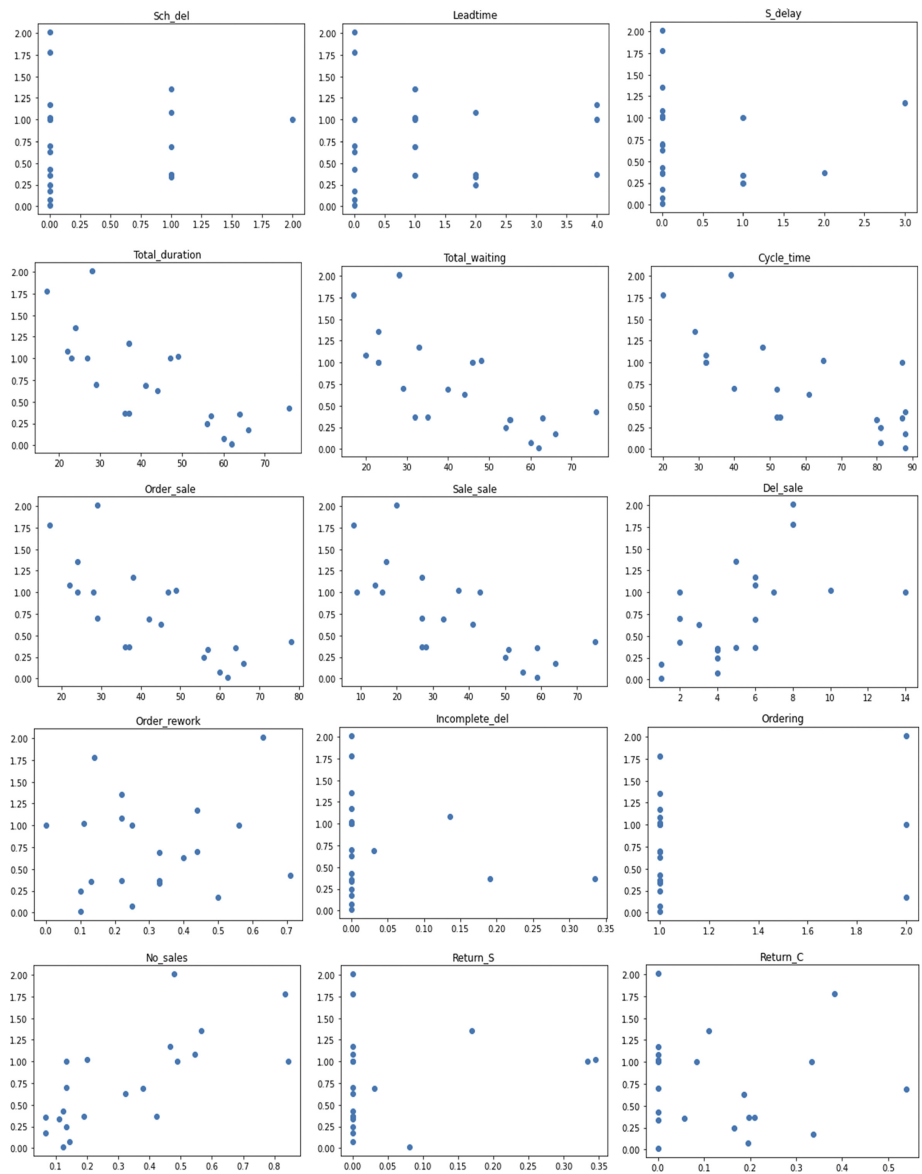


Figure 15.
Scatter plots of relative
efficiencies based on
each PI

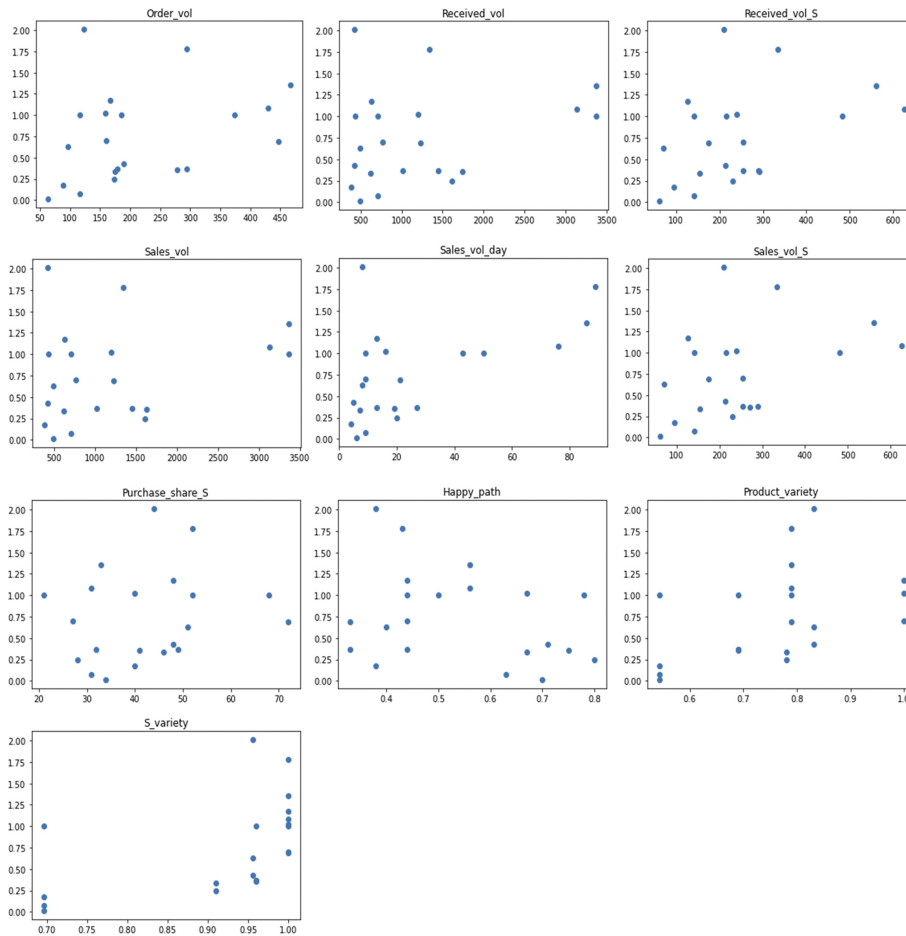


Figure 15.

in this evaluation. Such a detailed knowledge discovery with the existing raw data in an information system, using other approaches, would only be possible over a much longer time. However, the downside of PM was its disability to present a single multi-dimensional comparison on plentiful business units. This was why a comprehensive framework was designed, in which all the key elements of the most detailed discovery, a decent evaluation and comparison and an efficient way of suggesting enhancement points were contained.

Furthermore, the PCA was able to establish an appropriate link between the PM and the DEA to prevent any PI from being ignored, so if the required raw data is available, there is no limit for considering any kind of PI in this framework. Indeed, the last step of the framework will tell you which PIs were the most important ones, so there is no need for any filtering. It is evident that the overall performance and reputation of an organization depends on its every single unit, and also improving all of them in every performance aspect at the same time is not practical. In order to prioritize the improvements, the proposed framework can be useful. An effective performance improvement can benefit all the

stakeholders of an organization, from investors and managers (because of the financial resources allocated to the improvement opportunities which have the most impact on higher efficiency) to customers (which will receive more desirable services at each business unit such as brand variety).

The value of the case study in this paper was highly descriptive because the conceptual framework presented was applied simultaneously on real-life data of a chain store, and barriers were removed and improved practically. In the case study, the most notable result was the significant effect of time PIs on the resulted efficiency scores of distribution units; throughput times, amount of sales and flexibility (brand and supplier variety) were the most correlated ones to the relative efficiency of each unit. Thus, when the management level of this organization deciding on how much resource (financial, time, human, etc.) to allocate on each of the units in each aspect of performance, the challenge is specifically solved based on their data.

6. Discussion

Among the research that has been done in the PM field, there is a small number focused on performance evaluation of any type of organization, and for this field of study, whose one of the three techniques is named enhancement, this can be mentioned as a gap. Considering all the numerous organizations newly inclined to apply PM for their digitalized processes, a major action in the design of PM software can be taken. With the growing amount of data being generated in every day's processes of an organization, it is not logical to make decisions subjectively anymore, and even the smallest steps can be taken based on the available data. PM has been helpful and time-saving when organizations tend to monitor their PIs, but making decisions upon them is what has been left to human analysis so far. The problem is that not every unit will work perfectly on every aspect, and evaluating them, especially when for example the number of units is more than a hundred, is impossible. On the other hand, data science has been a help of human limitations for this purpose. Therefore, in the author's view, the combination of PM, PCA, DEA and data visualization has been a perfect match and the results would be more remarkable as the number of units and PIs grow dramatically.

7. Conclusions and future study suggestions

Competitive conditions have forced organizations' decision-makers to fully manage their units from top suppliers to end-users and to use strategic and operational tools to improve their overall performance. In this regard, considering the downsides of PM (disability to present a multi-aspect comparison), and DEA (time-consuming PIs' calculation), a comprehensive framework to evaluating the performance of business units from a process perspective was proposed as the primary purpose of the paper. It should be highlighted that by summarizing PIs using PCA, there is no need to select a limited number of PIs. While keeping many details involved, the evaluation can be quite continuous and agile, managers can quickly control everything and respond to any deficiencies as soon as possible by importing data from any new unit, new product, new supplier, new timeframe, etc.

The framework was implemented on product flow data using the integrated information system of an Iranian chain store. So far, few studies have used PM to evaluate performance; however, efforts have been made to measure the performance of different manufacturing processes, different paths of one process, students, jobs, emergency rooms and HIS systems, which evaluations were usually performed with a limited number of PIs. Thus, the contribution of this study is to provide the mentioned framework for a process-oriented comparison and improvement of different units in an organization. It is also concluded that

time and then flexibility are the most related criteria to the efficiency of a business unit, so the focus of enhancements to reach higher ranks is to escalate these two aspects.

In this paper, due to the limitation of data access in the studied company, a limited number of business units and PIs were analyzed; however, in the author's view, numerous units and PIs, enabling the target prediction and other data science techniques, could demonstrate the effectiveness of the framework better and allow it develop. It is recommended to consider cost PIs in future researches. Additionally, this evaluation was from the standpoint of different units of an organization; for future researches, the evaluation of different suppliers or product brands can be carried out, which results could highlight unreliable suppliers who increase process throughput times, brands whose orders are often problematic or any other useful data-based insights.

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Table A1.
Component matrix
(input PIs)

Component					
PIs	1	2	3	4	
X1	−0.562	0.499	0.037		0.160
X2	−0.440	0.820	0.246		−0.136
X3	−0.227	0.638	0.486		−0.221
X4	0.951	0.207	0.084		−0.041
X5	0.967	0.136	0.062		−0.029
X6	0.947	0.245	−0.044		−0.016
X7	0.948	0.198	0.100		−0.046
X8	0.983	0.112	0.086		0.020
X10	−0.738	0.072	−0.050		−0.325
X11	0.028	−0.445	0.793		0.041
X12	−0.299	0.516	0.204		0.511
X13	−0.010	−0.676	0.407		0.067
X14	−0.881	−0.153	−0.055		−0.048
X15	0.112	0.054	−0.741		−0.342
X16	−0.007	0.041	−0.391		0.751

Table A2.
Component matrix
(output PIs)

Component			
PIs	1	2	
X17	0.869		0.214
X18	0.964		−0.174
X19	0.957		−0.053
X20	0.965		−0.171
X21	0.805		0.113
X22	0.956		−0.048
X23	−0.423		0.602
X24	0.257		−0.663
X25	0.295		0.628
X26	0.598		0.656

Appendix 2
DEA-Solver input data

Performance
management
using process
mining

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Store	(I) Y1	(I) Y2	(I) Y3	(I) Y4	(O) Y5	(O) Y6
1	0.705199	0.558319	0	0.540827	0.888928	0.192946
2	0.44312	0.508439	0.417955	1	0.34257	1
3	0.166603	0.535924	0.582135	0.586436	0.992515	0.440839
4	0.233449	0.409612	0.313196	0.443335	1	0.465113
5	0.122849	0.238783	0.271736	0.73871	0.546648	0.773588
6	0.353418	0.658778	0.569464	0.849451	0.447725	0.48302
7	0	0.740251	0.647872	0.248353	0.243954	0.626232
8	0.311754	1	0.905614	0.929795	0.280245	0.662687
9	0.839846	0.57508	0.548108	0.430112	0.459531	0.323267
10	0.683913	0.710703	0.786549	0.364747	0.193949	0.464303
11	0.712277	0.680356	0.580171	0.467247	0.404633	0.197946
12	0.868454	0.51167	0.447224	0.400089	0.01986	0
13	0.34125	0.746947	1	0.114115	0.208054	0.858294
14	0.577973	0.518637	0.157499	0	0.374467	0.608956
15	0.461711	0.258735	0.651027	0.500017	0.331033	0.685182
16	0.599384	0.341042	0.587871	0.640458	0.086713	0.745882
17	1	0.423111	0.925461	0.445301	0.232663	0.534186
18	0.323818	0.011834	0.894335	0.39837	0.189494	0.705938
19	0.914359	0.254781	0.818736	0.870606	0	0.259055
20	0.274532	0	0.844009	0.424916	0.059589	0.413542
21	0.795789	0.467366	0.534997	0.603635	0.117336	0.020186

Table A3.
DEA-Solver input data

Appendix 3
Spearman's correlations

PIs	Correlation PI- efficiency	PIs	Correlation PI- efficiency
Average scheduled delivery duration	0.08	% Cases with return(S)	0.15
Lead-time (order-to-delivery)	0.08	% Cases with customer return	−0.17
Average supplier delay	−0.07	Average order volume	0.39
Average total process duration	−0.76	Overall received volume	0.19
Average total duration without waiting time	−0.77	Average received volume per-supplier	0.43
Maximum chain-cycle time	−0.77	Overall sales volume	0.19
Average order-to-sales duration	−0.75	Average sales volume per day	0.51
Average first-to-last sales duration	−0.78	Average sales volume per-supplier	0.43
Average delivery-to-sales(first) duration	0.65	Maximum purchase share among suppliers	0.22
% ordering-rework	0.13	% happy-path coverage	−0.27
% incomplete-delivery	−0.04	% brand-variety	0.63
13. Average ordering times to different suppliers	0.11	% supplier-variety	0.68
% no-sales-days	0.76		

Table A4.
Spearman's correlation
between variables